Object Oriented Programming (CSEG1044)
Unit 06: Capstone Project — Phase 1

Requirements and Use Cases

Project: Cafeteria Order and Billing System

1. Actors

The system has two primary actors and one supporting actor:

Actor	Type	Description
Admin	Primary Actor	Manages the system — adds/edits menu items, views sales reports, and manages user accounts.
Operator	Primary Actor	Front-desk staff who takes customer orders, generates bills, and views order history.
Database (MySQL)	Supporting Actor	Stores all persistent data including users, menu items, orders, and bills. Accessed via JDBC.

2. Use Case List

ID	Use Case	Actor(s)	Priority
UC-01	Login	Admin, Operator	**High**
UC-02	Manage Menu Items	Admin	**High**
UC-03	Place Order	Operator	**High**
UC-04	Generate Bill	Operator	**High**
UC-05	View Order History	Admin, Operator	**Medium**
UC-06	View Daily Sales Summary	Admin	**Medium**
UC-07	Logout	Admin, Operator	**Low**

3. Detailed Use Cases

UC-01: Login	
Use Case ID	UC-01
Use Case Name	Login
Actor(s)	Admin, Operator
Description	A registered user logs into the system using their username and password to access role-specific features.

Preconditions	The user has a valid account in the database.
Main Flow	1. User opens the application and the login screen appears. 2. User enters their username and password. 3. User clicks the Login button. 4. System validates credentials against the database. 5. System identifies the user role (Admin or Operator). 6. System redirects the user to the appropriate dashboard.
Alternate Flow	If credentials are invalid, system displays an error message. After 3 failed attempts, the login button is temporarily disabled.
Postconditions	User is authenticated and redirected to their role-specific dashboard.
Acceptance Criteria	✓ Valid credentials must grant access within 2 seconds. ✓ Invalid credentials must show a clear error message. ✓ Admin and Operator must be redirected to different dashboards. ✓ Password field must mask input characters.

UC-02: Manage Menu Items	
Use Case ID	UC-02
Use Case Name	Manage Menu Items
Actor(s)	Admin
Description	Admin can add new menu items, edit existing ones, or remove items that are no longer available.
Preconditions	Admin is logged in and on the Menu Management screen.
Main Flow	1. Admin navigates to the Menu Management section. 2. System displays the current list of menu items. 3. Admin clicks Add New Item and fills in item name, category, and price. 4. Admin clicks Save. System stores the new item in the database. 5. To edit: Admin selects an item, updates fields, and clicks Update. 6. To delete: Admin selects an item and clicks Delete. System confirms before removing.
Alternate Flow	If required fields are left empty, system highlights them and prevents saving. If price is non-numeric, system shows a validation error.
Postconditions	Menu item is added, updated, or removed from the database and the list refreshes.
Acceptance Criteria	✓ All fields (name, category, price) must be validated before saving. ✓ Item name must be unique — duplicates must be rejected.

	✓ Deleted items must no longer appear in the Operator's menu view. ✓ Price must only accept positive numeric values.

UC-03: Place Order	
Use Case ID	UC-03
Use Case Name	Place Order
Actor(s)	Operator
Description	Operator selects items from the menu, sets quantities, and places an order for a customer.
Preconditions	Operator is logged in. At least one menu item exists in the database.
Main Flow	1. Operator opens the Place Order screen. 2. System loads and displays all available menu items. 3. Operator selects items and sets quantity for each. 4. System shows a running subtotal as items are added. 5. Operator enters or selects the customer name. 6. Operator clicks Place Order. 7. System creates an Order and linked OrderItems in the database.
Alternate Flow	If no items are selected, system shows a warning. If quantity is zero or negative, system rejects the entry.
Postconditions	Order is saved to the database with a unique Order ID and timestamp.
Acceptance Criteria	✓ At least one item must be selected before placing an order. ✓ Quantity must be a positive integer. ✓ Order must be assigned a unique Order ID automatically. ✓ Subtotal must update in real-time as items are added or removed.

UC-04: Generate Bill	
Use Case ID	UC-04
Use Case Name	Generate Bill
Actor(s)	Operator
Description	Operator generates a bill for a placed order, showing itemised costs, tax, and the final payable amount.
Preconditions	An order has been placed and has not yet been billed.
Main Flow	1. Operator selects a completed order from the list. 2. Operator clicks Generate Bill.

	3. System retrieves all OrderItems linked to the order. 4. System calculates subtotal, applies tax (e.g., 5% GST), and calculates total. 5. System displays the bill with item breakdown. 6. Operator confirms and the bill is saved to the database.
Alternate Flow	If the order was already billed, system shows a warning and prevents duplicate bill generation.
Postconditions	Bill is saved with Bill ID, timestamp, total amount, and linked Order ID.
Acceptance Criteria	✓ Bill must show each item name, quantity, unit price, and line total. ✓ Tax must be calculated separately and shown on the bill. ✓ Final total must equal subtotal plus tax. ✓ Duplicate billing of the same order must be prevented. ✓ Bill calculation logic must not reside in the UI class.

UC-05: View Order History	
Use Case ID	UC-05
Use Case Name	View Order History
Actor(s)	Admin, Operator
Description	Users can view a list of all past orders with details such as order date, customer name, and total amount.
Preconditions	User is logged in. At least one order exists in the database.
Main Flow	1. User navigates to the Order History section. 2. System fetches all orders from the database. 3. System displays the list sorted by most recent first. 4. User can click on an order to view its full item breakdown.
Alternate Flow	If no orders exist, system displays a message: 'No orders found.'
Postconditions	Order history is displayed. No data is modified.
Acceptance Criteria	✓ Order list must show Order ID, customer name, date, and total. ✓ Orders must be sorted with the most recent at the top. ✓ Clicking an order must show its full item-by-item breakdown. ✓ Operator can only view orders; Admin can additionally filter by date.

UC-06: View Daily Sales Summary	
Use Case ID	UC-06
Use Case Name	View Daily Sales Summary

Actor(s)	Admin
Description	Admin views a summary of all orders placed on a selected date, including total revenue and number of orders.
Preconditions	Admin is logged in. Orders exist in the database for the selected date.
Main Flow	1. Admin navigates to the Daily Sales Summary section. 2. Admin selects a date using the date picker. 3. System queries the database for all orders on that date. 4. System calculates total number of orders and total revenue. 5. System displays the summary along with individual order entries.
Alternate Flow	If no orders exist for the selected date, system shows 'No sales recorded for this date.'
Postconditions	Summary is displayed. No data is modified.
Acceptance Criteria	✓ Admin must be able to select any past date. ✓ Summary must show total orders count and total revenue for the day. ✓ Revenue must match the sum of all bill totals for that date. ✓ Individual order rows must be listed below the summary.

UC-07: Logout	
Use Case ID	UC-07
Use Case Name	Logout
Actor(s)	Admin, Operator
Description	Logged-in user ends their session and is returned to the login screen.
Preconditions	User is logged in.
Main Flow	1. User clicks the Logout button. 2. System clears the session data. 3. System redirects the user to the login screen.
Alternate Flow	If user has an unsaved order in progress, system prompts them to confirm before logging out.
Postconditions	Session is terminated. Login screen is displayed.
Acceptance Criteria	✓ Clicking Logout must always return the user to the login screen. ✓ After logout, navigating back must not restore the previous session. ✓ If an order is in progress, a confirmation dialog must appear.

4. Non-Functional Requirements

Category	Requirement
Performance	Login and menu loading must complete within 2 seconds on a standard machine.
Usability	All screens must be navigable without a manual; labels must be clear and in plain English.
Security	Passwords must not be stored as plain text. Role-based access must be enforced at every screen.
Maintainability	Business logic (e.g., bill calculation) must be in separate classes, not inside Swing UI code.
Reliability	All database operations must use try-catch with JDBC; the app must not crash on SQL errors.
Portability	The application must run on any machine with JDK 11+ and a MySQL server installed.